

ABEL BIJU GEORGE

Kerala, India | +91 73067 85899

abelbijugeorge@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Python-focused software and AI/ML developer with hands-on experience building data-driven applications, machine learning workflows, backend APIs, and validation pipelines. Experienced in Python, Pandas, NumPy, Scikit-learn, REST APIs, PostgreSQL, Git/GitHub, debugging, data preprocessing, feature engineering, model evaluation, and visualization. Strong foundation in developing, validating, troubleshooting, and improving software and ML workflows, with practical experience working across structured and unstructured datasets and end-to-end application development.

EDUCATION

B.Tech in Information Technology, Amal Jyothi College of Engineering
CGPA: 6.96

May 2026

SKILLS

Programming: Python, JavaScript, Java, C

Version Control: Git, GitHub

AI/ML: Machine Learning, Generative AI, Scikit-learn, XGBoost, OpenCV, YOLOv8, Feature Engineering, Model Evaluation

Data: Pandas, NumPy, SQL, Data Cleaning, EDA, Dataset Validation, Data Transformation, Matplotlib, Seaborn

Backend: FastAPI, REST APIs, PostgreSQL, MySQL, MSSQL

Engineering: Software Development, Debugging, API Workflows, Data Pipelines, Problem Solving

Tools: MLflow, Roboflow, n8n **Strengths:** Analytical Problem Solving, Troubleshooting, Rapid Learning, Teamwork, Ownership

EXPERIENCE

AI Intern, Fitwell Tech

Jan 2026 – Apr 2026

- Prepared, annotated, validated, and analyzed real-world invoice datasets for computer vision and information extraction workflows.
- Used Roboflow and YOLO-based workflows to structure image datasets, validate annotations, and support machine learning experimentation.
- Investigated data-quality and preprocessing issues, iterating on validation workflows to improve the reliability of downstream ML outputs.

Software Developer Intern, United Business Group

Jun 2025 – Jul 2025

- Developed and integrated application workflows using .NET, Razor Pages, MSSQL, and backend APIs, working across application and relational-data layers.
- Translated requirements into working software features and debugged application logic, database interactions, API workflows, and data-related issues.

Data Analyst Intern, Fitwell Tech

Jun 2024 – Jul 2024

- Developed Python workflows to extract, clean, transform, validate, and structure information from raw JSON data using NumPy and regular expressions.
- Converted unstructured data into analysis-ready datasets for downstream processing, analysis, and reporting.

PROJECTS

TERRA – Personalized Tourism Recommendation System

[GitHub](#) | 2026

Python, Pandas, NumPy, Scikit-learn, Streamlit, Machine Learning

- Built an end-to-end tourism recommendation platform that transforms traveler preferences and tourism data into personalized destination recommendations and diverse trip shortlists.
- Developed machine learning workflows for attraction rating regression and visit-mode classification, with feature engineering, held-out evaluation, and recommendation scoring.
- Designed a diversity-aware recommendation engine with explainable match signals, enabling users to understand why destinations are recommended and build personalized trips.
- Deployed the system as an interactive Streamlit web application with traveler onboarding, recommendations, trip planning, model evaluation, data-quality diagnostics, and tourism insights.

EMIPredict-AI – Financial Risk & EMI Analytics

[GitHub](#) | 2026

Python, Pandas, NumPy, Scikit-learn, XGBoost, MLflow, Matplotlib

- Engineered and validated features from a dataset containing 400k+ records, creating financial indicators including EMI-to-income ratio, disposable income, expense-to-income ratio, and emergency-fund coverage.
- Built end-to-end machine learning pipelines using multiple input features for loan eligibility classification and EMI regression.
- Registered models with MLflow and implemented reproducible preprocessing, training, evaluation, and model-management workflows.

SolveStack – AI Problem Discovery Platform

[GitHub](#) | 2026

Python, FastAPI, PostgreSQL, React.js

- Built an end-to-end Python application that transforms unstructured online discussions into structured, searchable problem opportunities.
- Developed backend APIs and processing workflows for data collection, classification, tagging, filtering, and organization across multiple sources.
- Designed PostgreSQL data workflows and integrated backend and frontend components while independently debugging application and data-processing issues.

DefendHer – AI Public Safety System

[GitHub](#) | 2024

Python, OpenCV, YOLOv5

- Led development of a real-time computer vision system for detecting potential weapons and distress gestures from CCTV feeds.
- Designed and evaluated computer vision workflows for real-time inference and practical deployment scenarios.
- **Smart India Hackathon 2024 Grand Finale Finalist – Top 6 nationally.**

Zomato Restaurant Segmentation – Data Analysis Project

[GitHub](#) | 2026

Python, Pandas, NumPy, Scikit-learn, DBSCAN, Matplotlib, Seaborn

- Processed restaurant and review datasets containing 100+ restaurants and 10,000+ customer reviews, performing data cleaning, transformation, aggregation, and feature engineering.
- Built analytical features including average rating, review count, picture count, cost, and cuisine attributes for downstream machine learning analysis.
- Applied DBSCAN clustering and evaluated segmentation quality using Silhouette Score and Davies–Bouldin Index.

LEADERSHIP & ACHIEVEMENTS

- **President, init() Association**, Amal Jyothi College of Engineering (2025–26), coordinating technical initiatives, student teams, events, and communication with faculty and student stakeholders.
- **Smart India Hackathon 2024 Grand Finale Finalist** – Top 6 nationally; led the DefendHer team from problem identification through solution development and final presentation.
- **YIP 5.0 State Winner** – Kerala for FlexiCare, an innovative wheelchair-to-stretcher solution recognized at district and state levels.
- **2nd Runner-Up – Hack For Impact Hackathon**; conducted workshops on Prompt Engineering, AI, and UI/UX.